



KLE Technological
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**School of
Electronics and Communication Engineering**

**Research Experience for Undergraduates
on**

**Zero-Shot Scene Affordance Detection Using
Semantic Embeddings for Open-World
Perception**

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**SCHOOL OF ELECTRONICS AND COMMUNICATION
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This is to certify that the project entitled **“Zero-Shot Scene Affordance Detection Using Semantic Embeddings for Open-World Perception”** is a bonafide work carried out by the student **Tushar S Pyati (01FE22BEC177)**. The project report has been approved as it satisfies the requirements with respect to the research experience for undergraduate work prescribed by the university curriculum for **B.E. (VII Semester)** in the School of Electronics and Communication Engineering of **KLE Technological University** for the academic year **2025–2026**.

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ABSTRACT

Conventional scene understanding methods focus on semantic object categories, which often cannot capture the actionable usability of road environments. Visual affordance perception provides an action-centric alternative, identifying the regions that support or limit the agent behavior, such as driving etc,. Recent vision-language models allow for prompt-driven localization of affordances, however, their reliability under adverse weather and lighting conditions is insufficiently studied to motivate affordance-focused prompting and systematic robustness evaluation.

A pre-trained vision-language segmentation model, CLIPSeg, localizes actionable affordances within road scenes by using affordance-specific textual prompts. Multiple prompts are aggregated per category to improve spatial consistency. The proposed method is evaluated in both normal and adverse weather conditions in order to analyze affordance localization accuracy and robustness.

The proposed affordance-focused prompting approach achieves a mean IoU of 61.9% for drivable region localization, with good alignment between textual prompts and actionable road surfaces. The evaluation under diverse environmental conditions shows that drivable affordances remain comparatively stable despite visual degradation. These results demonstrate the suitability of vision-language models for actionable scene understanding while motivating further robustness analysis.

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Chapter 1

Introduction

Reliable perception of road environments is fundamental to autonomous driving systems, as errors at the perception stage can critically affect downstream decision-making and control. Most existing perception approaches are designed around semantic object recognition, emphasizing the detection and classification of entities such as vehicles, pedestrians, and traffic signs. While effective for scene understanding, such object-centric representations do not explicitly encode the actionable usability of the environment, particularly in terms of where an agent can safely drive.

Visual affordance perception provides an action-oriented alternative by identifying regions that support or restrict specific behaviors, such as driving. However, conventional affordance detection methods often depend on extensive supervised training and task-specific annotations, which limits their scalability and adaptability to open-world scenarios. Additionally, the robustness of affordance perception under adverse weather and lighting conditions remains underexplored, despite the prevalence of such conditions in real-world driving.

Recent advances in vision-language models enable prompt-driven, zero-shot localization of semantically meaningful regions, offering a promising direction for affordance-based perception without additional training. In this work, we investigate the use of affordance-focused textual prompts with a vision-language segmentation model to localize drivable regions in road scenes. Furthermore, we analyze robustness under adverse environmental conditions using Fourier-based preprocessing, providing insights into the stability of vision-language affordance perception in challenging real-world settings.

1.1 Motivation

Reliable perception of road environments is a critical requirement for autonomous and assisted driving systems, as perception failures directly propagate to downstream planning and control modules. Most existing perception approaches emphasize semantic object recognition, focusing on identifying discrete object categories such as vehicles, pedestrians, and traffic signs. While effective for scene interpretation, such object-centric representations often fail to directly encode the actionable usability of the environment, particularly in terms of where an agent can safely operate.

Visual affordance perception offers an action-centric alternative by explicitly identifying regions that enable or constrain specific behaviors, such as driving. However, traditional affordance detection methods typically rely on extensive task-specific annotations and supervised training, limiting their scalability and adaptability to open-world scenarios. Moreover, the reliability of

these methods under adverse weather and lighting conditions remains insufficiently explored, despite the prevalence of such conditions in real-world driving.

Recent advances in vision–language models provide an opportunity to address these challenges by enabling prompt-driven, zero-shot localization of actionable regions without additional training. Nevertheless, the extent to which such models can accurately and robustly capture drivable affordances across diverse environmental conditions remains unclear. This research is motivated by the need to systematically investigate affordance-focused prompting strategies and to evaluate their robustness under visual degradation, thereby advancing interpretable, flexible, and open-world perception for real-world driving applications.

1.2 Objectives

- To design affordance-focused textual prompts that guide the vision–language model toward identifying actionable scene categories (such as drivable, crossable, and obstructed regions) relevant for autonomous navigation.
- To achieve precise affordance localization by accurately detecting and segmenting the exact spatial regions in a scene that correspond to each actionable affordance.
- To evaluate the robustness of the affordance detection model under adverse weather and challenging lighting conditions (e.g., fog, rain, low illumination), analyzing its reliability for open-world perception scenarios.

1.3 Literature survey

Perception systems for autonomous driving have traditionally focused on semantic object detection and scene segmentation, where the goal is to recognize and localize predefined object categories. While such object-centric representations are effective for structured environments, they often fail to directly capture the actionability of a scene, such as whether a region supports driving. To address this limitation, affordance-based perception has emerged as an alternative paradigm that emphasizes functional regions enabling or constraining agent actions.

Early affordance detection approaches typically relied on supervised learning with task-specific annotations, limiting scalability and generalization to unseen environments. More recent research has explored open-vocabulary and open-world perception, enabling models to recognize previously unseen concepts using semantic embeddings. Vision–language models have played a central role in this shift by learning aligned representations between images and natural language, allowing semantic queries to be expressed through textual prompts. Such models have been successfully applied to zero-shot and open-vocabulary object detection and segmentation, demonstrating strong generalization across diverse categories.

Building on these representations, several works have investigated prompt-driven localization of affordances, where textual descriptions of actions are used to infer spatial regions of interaction. These approaches highlight the potential of pretrained vision–language models to ground functional concepts without explicit affordance supervision. However, existing studies also report performance degradation in challenging conditions, such as fog, rain, or low illumination, indicating sensitivity to appearance-level domain shifts.

To mitigate such domain gaps, domain adaptation techniques have been introduced, including methods operating in the frequency domain that selectively modify image appearance while

preserving structural content. These techniques have proven effective for analyzing robustness and improving generalization under adverse visual conditions. Despite these advances, a systematic evaluation of zero-shot affordance localization using vision–language segmentation models under diverse environmental conditions remains limited. This gap motivates further investigation into affordance-focused prompting strategies and robustness analysis for reliable open-world perception in real-world driving scenarios.

1.4 Problem statement

Existing scene understanding approaches are predominantly object-centric and operate under closed-set assumptions, limiting their applicability to open-world environments. Although semantic embeddings from vision–language models enable zero-shot reasoning, their capability to localize scene affordances at the pixel level remains underexplored. Additionally, the robustness of such zero-shot affordance detection under adverse visual conditions is not well understood. This work addresses the need for a scalable, zero-shot affordance localization framework for open-world perception.

Chapter 2

System design

This chapter outlines the architectural design of the proposed zero-shot scene affordance detection framework for open-world driving environments. The system is designed to localize actionable regions, specifically drivable areas, using semantic embeddings derived from vision-language models. The chapter details the functional components of the framework, including preprocessing, affordance prompting, segmentation, and output generation, and explains how these components interact to enable prompt-driven, training-free affordance localization. The design choices emphasize scalability, interpretability, and robustness to environmental variations, forming the foundation for the implementation and evaluation presented in subsequent chapters.

2.1 Functional Block Diagram

The functional block diagram represents the high-level organization and information flow of the proposed zero-shot drivable affordance detection system. It provides a modular view of the system, illustrating how different components interact to transform raw visual inputs into actionable affordance outputs. The design follows a sequential pipeline that emphasizes clarity, scalability, and interpretability.

The system begins with the input block, which receives RGB road scene images captured from a forward-facing camera. For scenes affected by adverse weather or lighting conditions, a preprocessing block based on Fourier Domain Adaptation (FDA) is employed. This block modifies the appearance statistics of the input image while preserving structural information, enabling robustness analysis under visual degradation without retraining the model.

The core processing block is the vision-language segmentation module, implemented using CLIPSeg. This block jointly processes visual features extracted from the image and semantic embeddings derived from affordance-specific textual prompts. By aligning visual and textual representations, the system enables prompt-driven, zero-shot localization of drivable regions.

The final block aggregates the segmentation outputs and produces a binary drivable affordance mask, which represents the navigable road surface. Overall, the functional block diagram highlights the integration of semantic reasoning and visual perception, demonstrating a systematic approach to open-world, action-centric scene understanding.

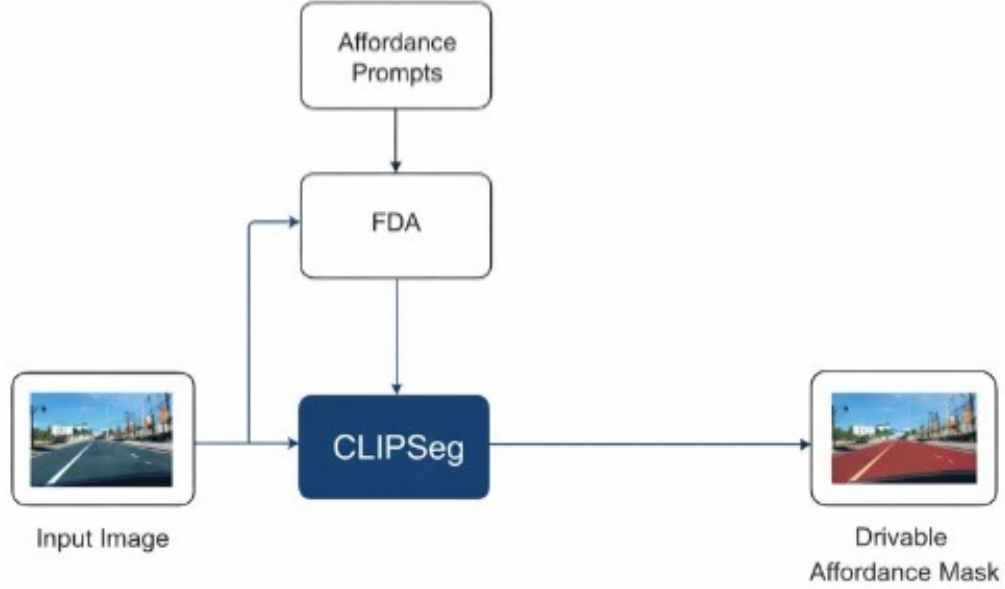


Figure 2.1: Functional Block diagram of Proposed System.

2.2 Design alternatives

The following design alternatives were considered for zero-shot drivable affordance detection:

Fully Supervised Semantic Segmentation Models: Conventional road segmentation models trained on annotated datasets can accurately detect drivable regions but require large-scale pixel-level supervision and exhibit limited generalization to unseen environments, making them unsuitable for open-world perception.

Object-Centric Open-Vocabulary Models: Open-vocabulary object detection and grounding-based approaches leverage semantic embeddings to recognize unseen classes; however, they focus on discrete objects rather than continuous actionable regions and therefore do not directly model drivable affordances.

2.3 Final design

The final design adopts a zero-shot, affordance-centric perception framework using a pretrained CLIPSeg model to enable prompt-driven, pixel-level localization of drivable regions. Affordance-specific textual prompts are encoded using the CLIP text encoder, while road scene images are processed through the CLIP image encoder. Multiple prompts are aggregated to improve spatial consistency and reduce linguistic bias.

To assess robustness under adverse weather and lighting conditions without retraining, Fourier Domain Adaptation (FDA) is used as a preprocessing step to adjust image appearance while preserving structure. The final output is a binary drivable affordance mask obtained through aggregation and thresholding, enabling zero-shot generalization and scalable open-world perception.

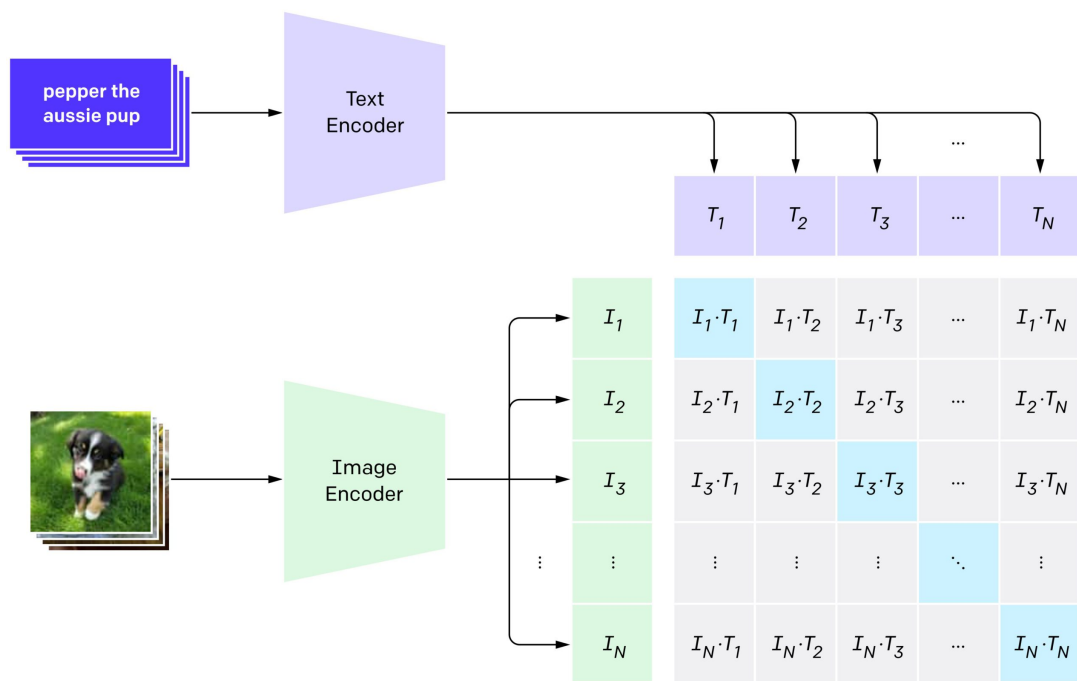


Figure 2.2: CLIP- architecture [1]

Chapter 3

Implementation details

This chapter describes the implementation aspects of the proposed zero-shot scene affordance detection framework. It details the system specifications, architectural components, and algorithmic procedures used to realize prompt-driven drivable affordance localization. The chapter also outlines the preprocessing strategy adopted for robustness analysis under adverse conditions and presents the datasets and evaluation metrics used to assess system performance. Together, these sections provide a comprehensive view of how the proposed design is translated into a practical and reproducible implementation.

3.1 Specifications and final system architecture

The proposed system is designed to perform zero-shot scene affordance detection with a focus on identifying drivable regions in open-world road environments. The architecture leverages a pretrained vision–language model to enable prompt-driven segmentation without task-specific retraining. The system operates on RGB images captured from forward-facing vehicle cameras and produces pixel-level affordance masks as output.

The core components of the system include an image preprocessing module, a vision–language embedding backbone, an affordance segmentation module, and an evaluation unit. For images captured under adverse weather conditions, Fourier Domain Adaptation (FDA) is applied as a preprocessing step to modify appearance-level statistics while preserving scene structure. This allows the system to analyze robustness under visual degradation without altering semantic content.

A pretrained CLIPSeg model is used as the main inference engine. The image encoder extracts dense visual features, while the text encoder processes affordance-specific prompts describing drivable regions. Multiple prompts are defined per affordance category to reduce linguistic bias and improve spatial consistency. The resulting segmentation masks are aggregated to produce a final drivable affordance map. Quantitative evaluation is performed on datasets with ground-truth annotations, while qualitative analysis is used for adverse-condition datasets.

3.2 Algorithm

Fourier Domain Adaptation is employed as a preprocessing algorithm to evaluate the robustness of the affordance detection framework under adverse weather and lighting conditions without retraining the model. FDA operates by modifying the low-frequency components of an image in the Fourier domain, which primarily encode global appearance information such as illumination, color distribution, and weather effects, while preserving high-frequency structural details.

Given a source image and a target image representing a different visual domain, both images are transformed into the frequency domain using the Fast Fourier Transform (FFT). The amplitude spectrum of the source image is partially replaced with the low-frequency amplitude components of the target image, while the phase spectrum of the source image is retained. The modified frequency representation is then transformed back to the spatial domain using the inverse FFT, producing an adapted image with altered appearance but preserved scene structure.

This FDA-processed image is subsequently passed to the CLIPSeg-based affordance segmentation pipeline. By isolating appearance-level changes, FDA enables systematic robustness analysis of the zero-shot affordance detection model under visually degraded conditions without altering semantic content or requiring additional training.

3.3 Dataset and Evaluation Metric

This section describes the datasets and evaluation criteria employed to assess the performance and robustness of the proposed zero-shot drivable affordance detection framework. The selected datasets are chosen to enable both quantitative accuracy measurement and qualitative robustness analysis under diverse environmental conditions. In addition, the evaluation metric is defined to objectively measure the spatial alignment between predicted affordance regions and ground-truth annotations, ensuring a consistent and interpretable assessment of drivable region localization performance.

3.3.1 Dataset

Two datasets are used in the implementation to evaluate both accuracy and robustness of the proposed system. The BDD100K dataset is used for quantitative evaluation, as it provides diverse driving scenes with ground-truth annotations for drivable regions under varying illumination and traffic conditions. This dataset enables objective measurement of affordance localization performance.

To assess robustness under adverse conditions, the DAWN dataset is employed. DAWN contains driving images captured under challenging weather conditions such as fog, rain, and haze. Since DAWN does not provide pixel-level ground-truth annotations for drivable affordances, it is used for qualitative evaluation and visual analysis of model stability under appearance degradation. FDA preprocessing is applied to these images to facilitate controlled robustness assessment.

3.3.2 Evaluation Metric

The performance of drivable affordance localization is evaluated using the Intersection over Union (IoU) metric. IoU measures the overlap between the predicted drivable region and the corresponding ground-truth annotation by computing the ratio of their intersection to their union. A higher IoU value indicates better spatial alignment and localization accuracy. IoU is computed only on datasets with available ground truth, while qualitative visualization is used for datasets lacking annotations.

Chapter 4

Optimization

This chapter discusses the optimization strategies adopted in the proposed zero-shot drivable affordance detection framework. Unlike conventional deep learning systems that rely heavily on training-time optimization, the proposed approach emphasizes inference-time and system-level optimization to improve localization quality, robustness, and computational efficiency. The chapter outlines the motivation for optimization, the types of optimization relevant to the system, and the rationale behind the selected optimization methods.

4.1 Introduction to optimization

Optimization in computer vision systems typically aims to improve performance metrics such as accuracy, robustness, and efficiency while minimizing computational overhead. In the context of the proposed framework, optimization does not involve training or fine-tuning model parameters, as the system operates in a zero-shot setting using pretrained vision-language models. Instead, optimization focuses on enhancing the quality of affordance localization through effective prompt design, output aggregation, threshold selection, and preprocessing strategies.

Given the open-world nature of road environments, the system must generalize across diverse scenes and adverse conditions. Therefore, optimization is directed toward stabilizing predictions, reducing noise in segmentation outputs, and ensuring consistent drivable region detection without compromising the zero-shot capability of the model.

4.2 Types of Optimization

Several types of optimization are relevant to the proposed affordance detection framework:

Prompt-Level Optimization: Multiple affordance-specific textual prompts are designed and aggregated to reduce linguistic ambiguity and improve alignment between semantic embeddings and visual features.

Output Aggregation Optimization: Segmentation masks generated from different prompts are combined using averaging and thresholding strategies to enhance spatial consistency and suppress spurious predictions.

Threshold Optimization: An optimal threshold is selected based on empirical evaluation to convert continuous probability maps into binary drivable affordance masks, balancing precision and recall.

Preprocessing Optimization: Fourier Domain Adaptation is applied selectively to images

captured under adverse conditions to modify appearance-level statistics while preserving scene structure, improving robustness without retraining.

Computational Optimization: The use of a single pretrained segmentation model and avoidance of retraining reduce computational cost and memory requirements.

4.3 Selection and justification of optimization method

Prompt-level optimization is selected as the primary optimization strategy in the proposed zero-shot drivable affordance detection framework due to its effectiveness, flexibility, and compatibility with vision-language models. Since the system operates without task-specific retraining, optimizing the textual prompts used to query the model becomes a critical factor in improving affordance localization performance.

Single textual descriptions often suffer from linguistic ambiguity and limited semantic coverage, which can lead to fragmented or inconsistent segmentation outputs. To address this, multiple semantically related prompts are defined for the drivable affordance category, capturing variations in phrasing such as road surface, drivable lane, and navigable path. These prompts are independently encoded using the text encoder and jointly evaluated against visual embeddings.

The segmentation outputs corresponding to each prompt are aggregated through averaging, followed by thresholding to generate a final drivable affordance mask. This aggregation strategy reduces sensitivity to individual prompt biases and enhances spatial consistency across the predicted regions. Prompt-level optimization thus enables fine-grained control over the semantic query space while preserving the zero-shot nature of the model.

The selection of prompt-level optimization is justified by its low computational overhead, absence of additional annotation requirements, and ability to improve robustness across diverse scenes and environmental conditions. This approach effectively enhances affordance localization quality while maintaining scalability and generalization in open-world perception settings.

Chapter 5

Results and discussions

This chapter presents the experimental results obtained from the proposed zero-shot scene affordance detection framework. Both quantitative and qualitative evaluations are conducted to assess the effectiveness of prompt-driven drivable affordance localization in open-world driving environments. Quantitative analysis is performed on datasets with available ground-truth annotations to measure localization accuracy using standard evaluation metrics. In addition, qualitative visual results are provided to examine spatial consistency and robustness under diverse environmental conditions, including adverse weather and lighting scenarios. The results highlight the ability of the vision-language-based segmentation approach to align semantic prompts with actionable road regions without task-specific training. Furthermore, the impact of prompt-level optimization and preprocessing strategies on affordance stability is analyzed, providing insights into the strengths and limitations of the proposed method in real-world perception settings.

5.1 Result Analysis

The proposed vision-language-based segmentation framework demonstrates effective localization of drivable regions across a wide range of real-world driving scenes. Quantitative evaluation conducted on the BDD100K dataset shows that the model achieves a mean Intersection over Union (IoU) of 0.619 at a threshold of 0.7, indicating a substantial spatial overlap between the predicted drivable regions and the corresponding ground-truth annotations. This level of overlap reflects the model’s ability to accurately capture the navigable road surface despite variations in road geometry, lane markings, and surrounding infrastructure.

The obtained performance highlights the effectiveness of affordance-focused prompting in guiding the segmentation model toward actionable regions rather than semantic object categories. The results suggest that the learned alignment between textual prompts and visual features enables consistent identification of drivable areas even in complex urban environments. Furthermore, the segmentation outputs exhibit strong spatial coherence, with predicted masks closely following the true road boundaries while effectively suppressing non-drivable regions such as sidewalks, buildings, and roadside structures.

Importantly, the observed performance remains stable across diverse scene layouts and illumination conditions, demonstrating the robustness of the zero-shot framework without reliance on task-specific training or fine-tuning. These findings indicate that the proposed approach can reliably infer drivable affordances using semantic embeddings alone, making it well-suited for open-world perception scenarios where exhaustive annotations and retraining are impractical.

Affordance	Mean IoU	Threshold
Drivable	0.619	0.7

Table 5.1: Average IoU for the drivable affordance.

5.2 Qualitative Visual Analysis

Qualitative results further highlight the spatial coherency of estimated drivable affordances. As revealed from the visual examples, the model offers accurate continuous road regions while effectively suppressing non-drivable areas such as sidewalks, buildings, and other surrounding structures. Even in visually complicated scenes, predicted masks show strong geometrical alignment with the true road, thus suggesting robust affordance localization.



Figure 5.1: Original image

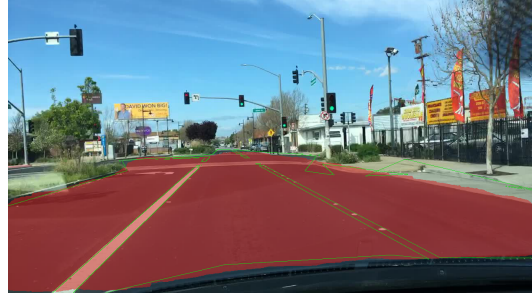


Figure 5.2: Drivable area detected



Figure 5.3: Original image

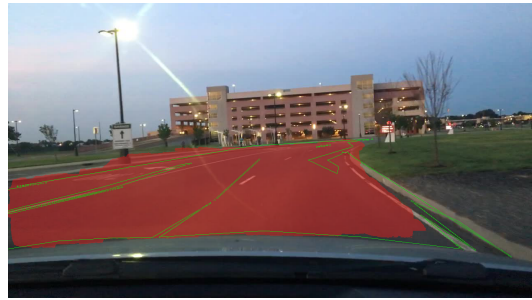


Figure 5.4: Drivable area detected

5.3 Qualitative Visual Analysis for Adverse weather

Qualitative analysis on the DAWN dataset shows that the model maintains coherent predictions of drivable regions under adverse weather and illumination conditions, such as fog and low-light conditions. Without any ground-truth annotations for quantitative evaluation, visual results demonstrate very stable and interpretable drivable affordance detection despite substantial appearance degradation. Collectively, these findings provide support for the effectiveness of the proposed approach in drivable affordance perception and emphasize its robustness in real-world driving environments.



Figure 5.5: Original foggy image



Figure 5.6: Drivable area detected



Figure 5.7: Original haze image



Figure 5.8: Drivable area detected



Figure 5.9: Original rainy image



Figure 5.10: Drivable area detected

Chapter 6

Conclusions and future scope

6.1 Conclusion

This work presents an affordance-aware perception framework to localize drivable regions by using a vision-language segmentation paradigm. The proposed approach achieves zero-shot identification of actionable road regions with affordance-focused textual prompts combined with a pretrained CLIP-based segmentation model without the need for task-specific training. Quantitative evaluations on the BDD100K dataset demonstrate very effective drivable affordance localization, reaching a mean IoU of 0.619.

Qualitative analysis on the DAWN dataset further demonstrates how the model can support coherent predictions even in adverse weather and light conditions. The Fourier Domain Adaptation is integrated to enable structured robustness analysis to disentangle appearance-level variations from the scene structure. All the results taken together thus indicate that the vision-language models yield a flexible and interpretable affordance-based scene understanding for open-world driving.

6.2 Future scope

Future work may extend this framework in a number of directions. First, the approach can be generalized to incorporate more affordances, such as crossable and obstructed regions, to further increase the accuracy of action-aware perception.

Second, incorporating temporal information from video sequences can improve the stability and continuity of affordance predictions in dynamic driving situations.

Third, further improvement of robustness is possible by combining Fourier-based adaptation with other domain-generalization techniques or by fine-tuning the vision-language model on datasets captured in adverse conditions.

Finally, coupling the affordance perception with downstream planning and control modules will enable end-to-end evaluation of its value for autonomous navigation performance in natural scenes.

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